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Construction Of a Large Artificial Island in a Major Offshore Brown Field

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Abstract

This paper presents a case study and guidance on technological solutions used to address the complexities encountered during the construction of an artificial island in a large offshore brownfield. It focuses on overcoming technical and logistical challenges related to transporting and installing massive volumes of construction materials—such as dredged sand and quarried rocks -amidst and atop live Oil & Gas assets, while maintaining the highest standards of Health, Safety, and Environment (HSE).

Keywords: Artificial Island, Offshore Brownfield, Hybrid Dredging, Floating Pipeline, MOTAS Coupling, Rock Logistics, Intermediate Stockpile, Marine Construction, Real-Time Monitoring, Sustainability

Introduction

The development of artificial islands has emerged as a strategic solution for expanding infrastructure in offshore Oil & Gas fields, particularly in regions with limited space and accessibility. These engineered landforms provide stable platforms for drilling, production, and logistics operations, while reducing the environmental footprint compared to traditional offshore structures.

Table 1—Abstract – Brownfield execution summary: scope, innovations, and results

SCOPE	INNOVATIONS	RESULTS
• 22 million Cu.m sand dredged • 6 million tons of rock transported • 40,000 Nos. precast elements transported	• DREDGING AND RECLAMATION ▪ Hybrid dredging strategy ▪ MOTAS quick couplings & Box anchors for 1.50 kilometer floating pipeline inside a live brownfield	• 3.2times Productivity using Large TSHD dredger
	• ROCK WORKS LOGISTICS ▪ Intermediate stockpile for rock materials ▪ Utilization large size flat top barges	• Approx. 20% increase in rock delivery efficiency with intermediate stock piling. • Continuity of rock material delivery.

Executing such a large-scale project amidst live Oil & Gas assets posed significant technical, logistical, and safety challenges. The proximity to critical infrastructure required innovative engineering solutions, precise coordination, and strict adherence to HSE standards. This paper presents a detailed case study of the technological strategies employed to overcome logistical and operational challenges, highlighting the integration of construction techniques and safety systems that enabled the successful delivery of new Artificial Island in Zakum oil field.



Figure 1—Island layout – 3D model

Project Overview

The scope of the new artificial island project included updating the detailed FEED design based on the final island size and location within the Lower Zakum Field. Key components involved:

- Execution of dredging and reclamation works.
- Soil improvement.
- Perimeter edge protection using rock material.
- Construction of a harbour breakwater.
- Installation of harbour marine structures, including a 600m concrete quay wall block and piled jetty structures.

Table 2—Volume of Works

Scope of Works	Quantities
Reclamation Works	22 Million Cu.m
Rock Works	6 Million Ton
Precasting Elements	40,000 Nos

Challenges Faced

Constructing artificial islands in brownfield areas presents unique challenges due to:

- Proximity to existing oil and gas facilities
- Presence of subsea assets
- Marine vessel draft limitations
- Adverse weather conditions
- Restricted site access (e.g., no nighttime vessel movement)

The challenges pertaining to two essential materials (sand and rocks) utilized in dredging and reclamation activities, as well as in rock material logistics, are comprehensively addressed in **sections 4.1 and 4.2**, respectively. Additionally, daily monitoring techniques for execution activities and the integration of advanced monitoring systems to safeguard marine operations are detailed in **Section 5.0 & lesson learnt detailed in Section 6**.

Dredging and Reclamation

Given the requirement to dredge 22 million cubic meters of sand from the borrow area and reclaiming the island within 20-month period in the brown field, the following challenges were encountered as listed in section 4.1.1.

Brownfield.

- Subsea assets: The subsea assets of the Zakum oilfield are primarily installed or buried at an approximate seabed level ranging from -6.00 to -25.00 meters. To mitigate the risk of damage to existing infrastructure, it is essential to select marine vessels that meet the under-keel clearance (UKC) requirements specific to the field, i.e., 0.50 times the vessel's draft.
- Marine vessel draft restriction: Limited vessel draft has impacted transport volume efficiency due to the minimum under keel clearance requirements imposed by ADNOC due to the presence of subsea assets in the field.
- Weather Limitation: Seasonal weather patterns have affected dredging and reclamation schedules, requiring operational adjustments.
- Dredging sand volume and reclamation duration: To dredge 22.0 million cubic meters over a 20 month period necessitates optimized planning and continuous operations.

Considering the above challenges, a comparative assessment has been conducted to evaluate baseline scenarios in terms of timeline, cost, and navigation hazards, with the objective of exploring the feasibility of deploying a small dredger versus a large dredger detailed in section 4.1.2 & 4.1.3.

Normal Scenario for Dredger Utilization.

Small Dredger: The use of smaller dredgers, considered due to draft limitations, led to longer operational cycles, and reduced overall efficiency. An additional cautious consideration when using small dredgers is the increased navigation hazards resulting from the considerable marine traffic in the field, which complicates coordination and operational cycles.

Operational and Technical Solution. To address the multifaceted challenges of constructing an artificial island within a live offshore brownfield, a hybrid dredging strategy was adopted, balancing operational efficiency, safety, and cost-effectiveness. The solution involved deploying a combination of large and small dredgers, strategically selected based on vessel draft limitations, under-keel clearance requirements, and site-specific constraints.

Optimized Dredger Deployment Strategy.

- *Large-Capacity Dredger Utilization:* To enhance reclamation efficiency, a high-capacity dredger (30,000 cubic meter hopper capacity) was mobilized. Its deployment significantly accelerated dredging operations, thereby shortening the overall reclamation timeline, and reducing exposure to seasonal weather-related disruptions. However, due to the vessel's draft (13.0 meter), under-keel clearance requirements had to be carefully considered, particularly near island boundaries where subsea assets are present.

As a result, the dredger was positioned approximately 1.5 kilometers away from the project boundary to discharge sand through a floating pipeline. Further technical details and solutions are elaborated in Section 4.1.3.2.

Large-Capacity Dredger

To implement the large-capacity dredger deployment strategy and carry out reclamation works using a 1.50 kilometer floating pipeline, comprehensive technical studies were conducted, followed by joint workshop sessions involving all stakeholders. Initially, the approach was not approved by the project end user due to safety concerns - primarily related to the dredger's capacity and draft - which limited its proximity to the project boundaries because of the presence of subsea assets in the field.

Further discussions were initiated to obtain approval for using a long floating long pipeline, hence the detailed risk assessment through workshops was conducted with end user's marine specialists to present the technology/method going to be in place to safeguard the floating pipelines as listed below with figures,

- Design of floating pipeline and installation:

The floating pipeline is engineered to remain buoyant on the sea surface, offering flexibility, mobility, and ease of adjustment to accommodate operational requirements and varying sea conditions. Its primary function is to transport dredged sand from the dredger to the designated fill area. The pipeline consists of 150-meter segments (10 strings), which are connected using MOTAS quick couplings. A key advantage of these couplings is their innovative connection mechanism. Unlike standard ball-joint or flanged connections, MOTAS couplings eliminate the need for personnel to be physically present on top of the floating pipes during the coupling process, significantly enhancing safety. The detailed mooring analysis was done to verify the safe operation due to weather conditions. For the mooring calculation of the floating pipeline, the specialized software 'OrcaFlex' has been used to determine the weather conditions under which the pipeline can remain in place, in our case maximum wave height 5.6m was considered.

To secure the floating pipeline in a fixed position, a gravity box anchor system, each anchor weighing 180 tons - was specifically designed and implemented. For the deployment of these 10 box anchors, a purpose-built box anchor pontoon was mobilized. The pontoon is equipped with an open deck and sea fastening locks, facilitating the safe transportation and efficient installation of the anchoring equipment at the project site.

- Performance of floating pipeline:

Although the floating pipeline was designed to withstand the specified weather criteria, its performance during pumping operations was continuously monitored. Key observations are summarized below:

Weather Limitation: Weather monitoring and forecasting are essential for ensuring the safe and efficient execution of dredging operations. Third-party weather forecasts are issued every 12 hours, providing detailed wind and wave predictions, which are disseminated to all operational vessels. Additionally, two Obscape wave buoys have been deployed to enable real-time monitoring of wave conditions.

The dredger crew and the Operational Superintendent routinely assess these forecasts, focusing on critical parameters such as wave height, wave direction, and wave period, in alignment with dredging plans.

Monitoring System: A safety corridor of 50 meters has been established on both sides of the floating pipeline to ensure secure operations and prevent unauthorized access. To monitor pipeline movement and safeguard ongoing activities, two escorting tugs have been deployed. The tug continuously patrols the corridor, providing real-time oversight and immediate response capability in case of any operational deviation or external interference.

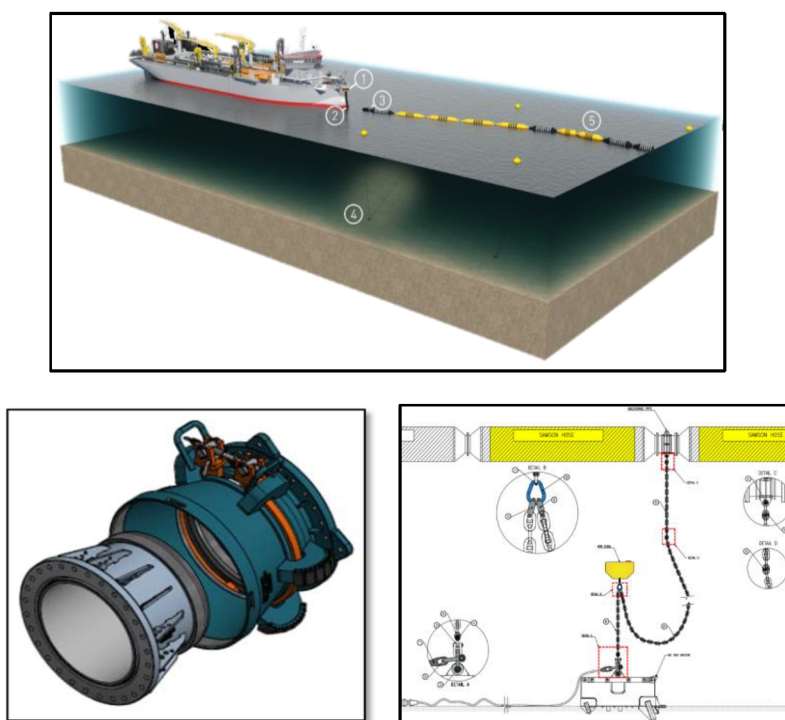


Figure 2—Floating pipeline arrangements; (Left) – MOTAS coupling; (Right) – Box anchoring system

Results and Achievements. The deployment of a large hopper dredger with the 1.5 kilometer floating pipeline resulted in a significant reduction of the execution time. Similar output would have been achieved by utilizing at least 3 smaller hopper dredgers. This scenario would have resulted in an increase in the hazards due to navigation in the field, reduced efficiency which leads to increased cost.

Conclusion. The successful construction of new Artificial Island within the Lower Zakum offshore brownfield exemplifies the effectiveness of integrating engineering solutions with stringent safety and environmental standards. By implementing a targeted dredging strategy and deploying a high-capacity hopper dredger supported by a floating pipeline system, the project overcame substantial logistical and technical challenges, including vessel draft limitations, proximity to subsea assets, and adverse weather conditions and as pointed out as below:

- a. A novel hybrid dredging strategy was implemented, combining large and small dredgers to optimize reclamation efficiency while ensuring safety near sensitive subsea infrastructure.
- b. The MOTAS coupling system and custom-designed box anchors represent a significant advancement in 1.5 kilometers long floating pipeline safety and operational efficiency.

The use of a large hopper dredger resulted in a **3.2-times** increase in productivity compared to a standard hopper dredger. This achievement - driven by optimized dredging operations, real-time marine

traffic coordination, proactive pipeline maintenance, and robust environmental safeguards - underscores the importance of collaborative planning, stakeholder engagement, and adaptive execution in complex offshore developments.

This case study on dredging and reclamation provides valuable insights and practical guidance for future artificial island projects in brownfield environments.

Rock Material Logistics

Rock material is one of the essential components in shore protection for artificial island construction. For this project, approximately 6.0 million tons of rock were transported from a quarry located in the Northern Emirates to the project site (Approx. distance of 140 nautical miles). This operation required the extensive deployment of rock barges, involving continuous loading, transit, and unloading cycles. The logistics operation encountered several significant challenges, which are outlined in the following sections:

- *Large number of barges:* To meet the rock delivery schedule, we mobilized 23 no. of combinations (tugs and barges) for rock material transportation. This includes five (5) large barges (> 20,000 tons capacity).
- *Brown field requirements:* Due to field operational constraints, barge movement was not allowed during dark hours.
- *Weather Limitation:* Seasonal weather conditions adversely impacted barge towing schedules, resulting in potential delays and the need for operational adjustments to maintain delivery timelines. The cumulative downtime was approximately 20-25%.
- *Delivery requirements as forecasted:* In line with project schedule and milestone targets, the delivery of rock materials required precise forecasting and scheduling. Ensuring the correct tonnage and material grade was delivered and placed at various construction locations, as per design specifications posed significant logistical and planning challenges.

Considering the above challenges, a comparative assessment was undertaken to evaluate the baseline scenario in terms of timeline, cost implications, and navigation hazards, against a solution-oriented approach involving the deployment of large-capacity barges ($\geq 10,000$ tons) and the use of intermediate stockpile locations for storage and loading, as detailed in Sections 4.2.1 and 4.2.2.

Operational and Technical Solution. To enhance operational efficiency, the solution involved deploying a combination of barges ranging from 10,000 to 25,000 tons in capacity, strategically selected based on site-specific constraints as detailed in Section 4.2.2.1. Additionally, the use of an intermediate stockpile at the Dubai Waterfront was considered, as outlined in Section 4.2.2.2.

Deployment of Large-Capacity Barges ($\geq 10,000$ Tons)

To reduce the frequency of barge movements and improve overall logistics efficiency, the deployment of large-capacity barges (>20,000 tons capacity) was considered. This approach offers the following advantages:

- *Reduced Marine Traffic:* Fewer trips are required to transport the same volume of rock material, minimizing congestion and navigation risks in the oilfield.
- *Lower Tug Dependency:* Larger barges reduce the need for multiple tug operations, streamlining towing logistics and lowering operational costs.
- *Improved Delivery Efficiency:* Higher payload capacity allows for better alignment with construction schedules and milestone targets.
- *Cost Optimization:* Although larger barges may have higher charter rates, the reduction in trip frequency, and associated fuel consumption and support operations contributes to overall cost savings.

Based on the proposed solution, the use of higher capacity barges as (>20,000-ton capacity barges) with combination of medium capacity barges was considered essential. All operational and field-specific requirements were successfully met, confirming the suitability of this approach for the project.

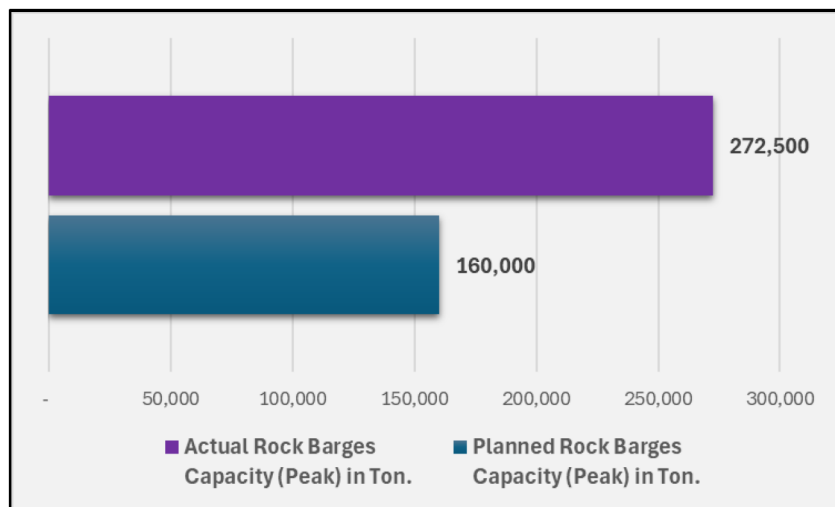


Figure 3—Improvement in cycle and Productivity using large capacity barges

Utilization of Intermediate Stockpile Locations

To address field access restrictions and enhance flexibility in material handling, intermediate stockpile locations were proposed. The selected stockpile site is located approximately 90 nautical miles from the quarry, within the Dubai Waterfront area, which also serves as the laydown zone for the project's main subcontractor.

These temporary storage locations offer the following benefits:

- *Decoupled Delivery and Placement:* Rock materials can be delivered in advance and stored at the stockpile. Barges can then be loaded from the stockpile and transported to the project site without interfering with other marine vessel movements and waiting for port clearance.
- *Reduced Waiting Times:* Barges can offload materials without delay, avoiding anchorage outside the field.
- *Enhanced Scheduling Flexibility:* Stockpiles enable better coordination with weather windows and construction sequencing.
- *Operational Resilience:* In case of unforeseen delays or disruptions, stockpiles provide a buffer to maintain continuous construction activity.

Results and Achievements. The implementation of the proposed operational and technical solutions resulted in measurable improvements across key performance indicators related to marine logistics and material handling:

- *Cycle Time Reduction:* The deployment of large-capacity barges (up to 25,000 tons) significantly reduced the number of cycles required for rock material transport. This led to a marked decrease in cycle time and improved overall delivery efficiency, as evidenced in Fig.4.
- *Increased Throughput and Productivity:* Higher payload capacity enabled better alignment with construction milestones, reducing delays and enhancing daily throughput. The combination of large and medium-capacity barges allowed flexible scheduling and optimized resource utilization.

- *Minimized Marine Congestion and Tug Dependency:* Fewer barge movements and reduced reliance on tug operations contributed to lower navigation risks and operational costs.
- *Improved Material Flow via Intermediate Stockpile:* The establishment of a stockpile at the Dubai Waterfront, approximately 90 nautical miles from the quarry, enabled decoupling of delivery and placement activities. This mitigated delays due to berth congestion and field access restrictions, as shown in Fig.5.
- *Enhanced Operational Resilience:* The stockpile provided a buffer against weather-related disruptions and unforeseen delays, ensuring continuity in rock placement operations and maintaining construction progress.

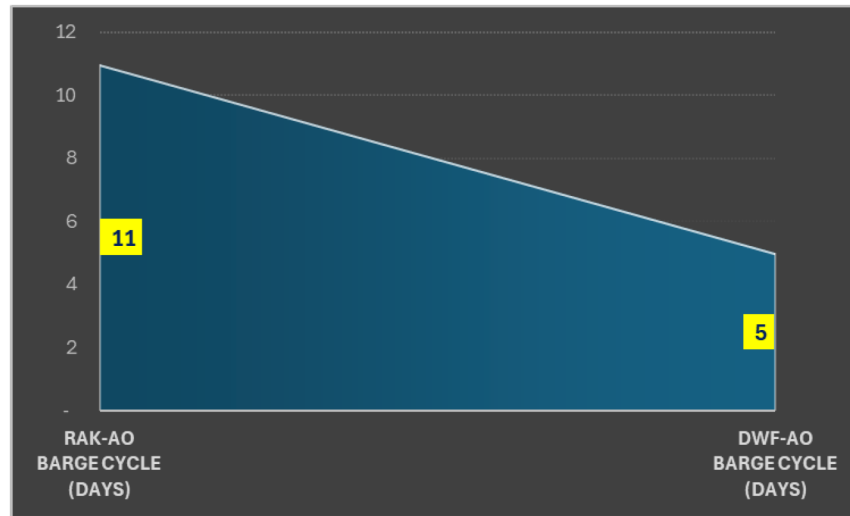


Figure 4—Utilization of Dubai Water Front yard vs Direct delivery

Conclusion

The integrated transport strategy, leveraging an intermediate stockpile with harbor access and high capacity barges, demonstrates a forward-thinking approach to logistics and sustainability. The strategic use of intermediate stockpiles enabled decoupling of delivery and placement operations, enhancing resilience and reducing marine congestion. By improving delivery efficiency by **20%**, the project not only accelerated material movement but also optimized resource utilization. The use of large barges allowed for an optimization of fuel consumption of nearly **30%** in comparison with smaller barges meeting the same transport capacity. This helped lower the carbon footprint and align with broader sustainability goals. This initiative serves as a benchmark for future similar infrastructure projects, showcasing how smart logistics planning and the adoption of scalable marine transport solutions can deliver measurable performance gains while supporting environmental responsibility.

Daily Monitoring - Survey Systems

Daily monitoring is a critical aspect of marine construction, particularly due to the sequential execution of rock material installation and sand reclamation activities. Seabed daily changes resulting from these operations must be closely tracked to ensure safe and efficient marine operations.

Advanced survey systems, such as Qinsy™ software, are deployed on the project to capture real-time data on seabed topography and material displacement. When integrated with RTK GNSS technology, Qinsy™ plays a vital role in underwater construction activities, including dredging, reclamation, rock works, and hydrographic surveying. These tools deliver precise 3D visualizations that significantly enhance situational awareness. The software is installed on all marine equipment and vessels operating on the

project. By integrating a comprehensive suite of survey technologies, including GNSS RTK, Echoscope, laser surveying and unmanned surface vehicles (USVs) construction teams can execute complex underwater tasks with greater accuracy and confidence. This technological synergy supports informed decision-making, minimizes operational risks, and optimizes workflow efficiency in challenging marine environments. Continuous monitoring enables early detection of anomalies, supports operational decisions, and ensures compliance with safety protocols in brownfield environments—contributing to the project's overall success throughout its duration.

Lessons Learned

The new Artificial Island project presented a unique set of challenges due to its location within a live offshore brownfield environment. The successful execution of this complex project yielded several key lessons that can serve future artificial island developments and offshore infrastructure projects in live brownfields:

- **Hybrid Dredging Strategy Enhances Operational Flexibility**
Deploying large hopper dredgers enabled the project team to balance productivity with safety. Large dredgers, supported by a 1.5-kilometer floating pipeline (the first of its kind in a live brownfield environment) accelerated bulk reclamation.
- **Innovative Pipeline Coupling and Anchoring Improves Safety**
The use of MOTAS quick couplings and custom-designed box anchors significantly improved the safety and efficiency of floating pipeline operations. Eliminating the need for manual coupling on floating pipes reduced personnel exposure and enhanced operational control, especially under variable weather conditions.
- **Intermediate Stockpiles Increase Scheduling Resilience**
Establishing a stockpile at the Dubai Waterfront decoupled material delivery from placement activities. This strategy mitigated delays caused by field access restrictions and berth congestion, enabling better alignment with construction milestones and weather windows.
- **Real-Time Monitoring Systems Support Decision-Making**
Advanced survey systems, including GNSS RTK and Qinsy™, provided continuous feedback on seabed conditions and material placement. These technologies enhanced situational awareness, reduced operational risks, and supported adaptive planning throughout the project lifecycle.
- **Stakeholder Engagement is Critical in Brownfield Environments**
Early and continuous engagement with field operators, marine specialists, and regulatory bodies was vital to securing approvals and aligning safety protocols. Collaborative workshops and joint risk assessments helped build consensus around innovative solutions such as the 1.5-kilometer long floating pipeline inside a live brownfield.
- **Scalable Logistics Solutions Drive Sustainability**
The deployment of large-capacity barges (>20,000Tons) and optimized transport cycles reduced fuel consumption and marine traffic, contributing to lower emissions and improved environmental performance. These scalable logistics strategies are applicable to other offshore projects seeking to balance efficiency with sustainability.